

7. (a) (i) A group of students were given three water samples. One was soft water, one was temporary hard water and one was permanent hard water.
- Describe a method that the students could use to find out which is which. Give the expected results. You do **not** need to include the detail required to ensure a fair test. [4]

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- (ii) The samples from part (i) were labelled **X**, **Y** and **Z** and sent to have their ion content measured. The table shows the results.

Ions present	Ion content in water sample (mg/dm ³)		
	X	Y	Z
sodium	29	116	23
potassium	12	15	11
magnesium	31	4	98
calcium	141	2	27
hydrogencarbonate	30	19	219
chloride	17	14	20
sulfate	346	6	27
nitrate	12	15	19

Use this information to find whether sample **Z** is the soft water, the temporary hard water or the permanent hard water. Give a reason for your choice. [2]

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- (b) (i) Permanent hard water can be softened by ion exchange. Explain how ion exchange works. [2]

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- (ii) Explain why the resin in ion exchange stops working after continued use. [1]

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- (c) A litre bottle of mineral water contains 184 mg of dissolved calcium sulfate, CaSO_4 . Calculate the number of moles of calcium sulfate present.

Give your answer to **three** significant figures.

$$1 \text{ mg} = 0.001 \text{ g} \quad [2]$$

$$A_r(\text{Ca}) = 40 \quad A_r(\text{S}) = 32 \quad A_r(\text{O}) = 16$$

Moles of calcium sulfate = mol

